A Modification of Davidon's Minimization Method to Accept Difference Approximations of Derivatives

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ABSTRACT. A modification of Davidon's method for the unconstrained minimization of a function of several variables is proposed in which the gradient vector is approximated by differences. The step sizes for the differencing are calculated from information available in the course of the minimization and are chosen to approximately balance off the effects of truncation error and cancellation error. Numerical results and comparisons with other methods are given.

1. *Introduction*

In recent years a number of new techniques for the unconstrained minimization of a function, Φ, of several variables have been published, notably in *The Computer Journal* [1, 3, 4, 6, 9]. The most successful of these methods are based on the properties of quadratic functions and are characterized by excellent convergence around the minimum. The ability to approach the minimum from points far removed varies from method to method; however, all of them are definitely superior to such standard techniques as steepest descent or random search.

The new techniques take as their starting point the quadratic function

$$\Phi(x) = \Phi_0 + x^T b + \tfrac{1}{2} x^T A x\,, \tag{1}$$

where Φ_0 is a scalar, $x^T = (\xi_1, \xi_2, \cdots, \xi_n)$ is a n-vector of variables, b is a constant n-vector and A is a symmetric, positive definite matrix. The contours of the function (1) are ellipsoids whose principal axes are the eigenvectors of A with lengths proportional to the reciprocals of the eigenvalues of A. The gradient vector of the function is

$$g(x) = b + Ax,$$

and its minimum is at the point $x_{\min} = -A^{-1}b$ for which $g(x_{\min}) = 0$.

The methods proceed by a sequence of linear minimizations. Starting with a point x_0, and a direction d_0, a number σ_0 is determined so that $\Phi(x_0 + \sigma_0 d_0)$ is a minimum along the line $x_0 + \sigma\, d_0$. A new direction d_1 is then chosen and the process is restarted from the point $x_1 = x_0 + \sigma_0 d_0$. It is evident that the directions d_i may be chosen in a variety of ways. For example, the method of steepest descent takes $d_i = -g(x_i)$. Most of the newer methods choose the d_i in some systematic manner so that if the function in question is the quadratic (1), the method will arrive at the minimum after a finite number of linear minimizations.

Methods which find the minimum or stationary point of a quadratic function in

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a finite number of steps have been referred to in the literature as quadratically convergent methods. Unfortunately this expression is also used to refer to the asymptotic behavior of iterations such as Newton's method for finding a root of an equation. Since the minimization methods discussed in this paper are iterative when applied to a general differentiable function, it is easy to confuse the two senses of the expression. In fact, quadratically convergent minimization methods generally exhibit excellent convergence from points near the minimum of a differentiable function, since most such functions may be well represented by (1) around their minima.

Minimization methods fall naturally into one of two categories: those which require the gradient vector to be evaluated and those which do not. Of the former, the method of Davidon [1, 3] is the most powerful. Ordinarily the evaluation of the gradient is performed by a closed subprogram which must be furnished by the user. In this paper we show how the components of the gradient may be estimated by difference quotients, thus making the method applicable to problems in which the gradient cannot be easily calculated. The heart of this modification is an algorithm, developed in Section 3, which uses information obtained in the course of the minimization to estimate optimal step sizes for differencing.

A few general comments on other methods are appropriate here. Fletcher and Reeves [4] have proposed a quadratically convergent method based on the method of conjugate gradients for solving a positive definite system of linear equations. In terms of function and gradient evaluations, this method is somewhat less efficient than Davidon's. But the mechanics of the method require fewer arithmetic operations, and it is easier to program.

Fletcher [2] has evaluated three minimization methods which do not require derivatives to be calculated: the method of Davies, Swann and Campey [9], the method of Powell [6] and a modification of a method by Smith [8]. His conclusions are as follows. Smith's method is noticeably inferior to the others both in terms of the number of linear minimizations and the number of function evaluations required. If the method of Davies, Swann and Campey, which is not quadratically convergent, is compared with Powell's method in terms of number of linear minimizations required, then they stand on about an equal footing, with the first method somewhat more efficient at points removed from the minimum. On the other hand, if the methods are compared in terms of number of function evaluations, Powell's method emerges as definitely superior. Also around the minimum the quadratic convergence of Powell's method asserts itself, and the final convergence to the minimum is more impressive.

Since the modification of Davidon's method proposed in this paper substitutes additional function evaluations for the calculation of a gradient, it should be compared with methods which do not require derivatives. Powell's method has been chosen as a standard of comparison, as it is the most efficient of these methods in terms of function evaluations.

2. *Davidon's Method*

The method which is modified in this paper was devised by Davidon and published in 1959 [1]. In 1962 Fletcher and Powell [3] gave a full exposition of the algorithm in a modified form stressing its quadratic convergence. What follows is a brief description of the method as set forth by Fletcher and Powell. Those interested in further details and proofs are referred to their paper, or to Davidon's original report.

The method proceeds by forming a sequence of positive definite matrices, $H_0, H_1, \cdots$. If the function to be minimized is the quadratic function (1), then $H_{n-1} = A^{-1}$, so that the minimum is given by $-H_{n-1}\, g(x)$. In the nonquadratic case the H_i approximate the inverse matrix of second derivatives of the function in question, and $-H_i\, g(x)$ may be expected to be a good direction along which to look for a minimum. A typical iteration goes as follows:

1. Given x_i and $g_i = g(x_i)$, calculate

$$d_i = -H_i g_i . \tag{2}$$

2. Find σ_i so that $\Phi(x_i + \sigma_i d_i)$ is a minimum along $x_i + \sigma\, d_i$.
3. Set $x_{i+1} = x_i + \sigma_i d_i$, and calculate $g_{i+1} = g(x_{i+1})$ and $y_i = g_{i+1} - g_i$.
4. Form H_{i+1} by

$$H_{i+1} = H_i + \sigma_i \frac{d_i d_i^T}{d_i^T y_i} - \frac{H_i y_i y_i^T H_i}{y_i^T H_i y_i} . \tag{3}$$

Any positive definite matrix may be used to start the iteration. In the absence of any specific information on the function to be minimized, H_0 is usually taken as the identity matrix. Fletcher and Powell suggest that the procedure be terminated when the components of both the predicted direction d_i and the actual step $\sigma_i d_i$ fall below a specified level.

In Section 3, where a method is developed for choosing step sizes to approximate derivatives by differences, it proves necessary to have estimates of the second derivatives of the function along the coordinate directions. These are provided by the diagonal elements of H_i^{-1}, which may be calculated recursively as follows. With the aid of (2), eq. (3) may be written in the form

$$H_{i+1} = H_i + \sigma_i \frac{H_i g_i g_i^T H_i}{d_i^T y_i} - \frac{H_i y_i y_i^T H_i}{y_i^T H_i y_i} .$$

If the modification rule given in Householder [5, p. 123] is applied first to

$$H_i' = H_i + \sigma_i \frac{H_i g_i g_i^T H_i}{d_i^T y_i}$$

and then to

$$H_{i+1} = H_i' - \frac{H_i y_i y_i^T H_i}{y_i^T H_i y_i} ,$$

the result is

$$H_{i+1}^{-1} = H_i^{-1} + \left(\frac{1}{\beta_i \sigma_i} - \frac{\rho_i}{\beta_i^2}\right) y_i y_i^T + \frac{1}{\beta_i} (g_i y_i^T + y_i g_i^T), \tag{4}$$

where $\rho_i = g_i^T d_i$ and $\beta_i = y_i^T d_i$. In fact (4) shows that any element of H_{i+1}^{-1} may be computed from the corresponding element of H_i^{-1} without any knowledge of the other elements of H_i^{-1}. In particular the diagonal elements of H_{i+1}^{-1} may be easily calculated from those of H_i^{-1}.

3. *Calculating Accurate Derivatives*

Davidon's method requires that a gradient vector be calculated in the course of each iteration, i.e., that the derivatives of the function be evaluated along the coordinate directions. The modification proposed in this paper is simply that the jth component of the gradient vector be estimated by

$$\gamma_j \simeq \frac{\Phi(x + \delta_j e_j) - \Phi(x)}{\delta_j}, \tag{5}$$

where e_j is the vector with its jth component unity and its other components zero. Equation (5) represents a simple difference scheme for the estimation of the derivative. Of course the use of a central difference approximation would yield better estimates of the derivatives; however, its general use in this method would require an additional function evaluation for each component of the gradient, thus impairing the efficiency of the method. Also numerical evidence indicates that, with appropriate choice of the δ_j, the approximation (5) is often quite adequate, and the following informal analysis is directed toward choosing a good δ_j. Nonetheless, at certain points in the iteration it is desirable to use central differences, as is shown in what follows.

In this section it is sufficient to consider the problem of approximating the derivative of a parabola by differences, since if Φ is the quadratic function (1), then along e_j

$$\Phi(x + \lambda e_j) = \Phi(x) + \gamma_j\lambda + \tfrac{1}{2}\alpha_{jj}\lambda^2. \tag{6}$$

Here γ_j is the jth component of the gradient at x and α_{jj} is the jth diagonal element of A. In step 3 of the Davidon method, where g_{i+1} must be evaluated, the parameters of (6) may be estimated from numbers calculated in the course of the iteration as follows: $\Phi(x)$ by $\Phi(x_{i+1})$, γ_j by $\gamma_j^{(i)}$ and α_{jj} by $\alpha_{jj}^{(i)}$, the jth diagonal element of H_i^{-1}.

Thus consider the problem of estimating $\gamma = \phi'(0)$ of the parabola

$$\phi(\lambda) = \phi_0 + \gamma\lambda + \tfrac{1}{2}\alpha\lambda^2$$

by

$$\gamma \simeq \frac{(\phi(\delta) - \phi_0)}{\delta} = \frac{\Delta\phi}{\delta}. \tag{7}$$

There are two major sources of error in this calculation: the truncation error due to the mathematical inadequacy of the approximation (7), and the error due to cancellation of significant figures when $\phi(\delta) - \phi_0$ is calculated on a machine with finite word length. The relative effect of the truncation error may be estimated by

$$\left|\frac{\Delta\phi/\delta - \gamma}{\gamma}\right| = \frac{1}{2}\frac{|\alpha\delta|}{|\gamma|}. \tag{8}$$

To get a bound for the second source of error requires that the accuracy to which ϕ is computed be known. Let the calculated value of ϕ be given by $\phi^* = \phi(1 + \epsilon)$, where $|\epsilon| \leq \eta$, a known error bound. Then

$$\begin{aligned} \phi_0^* &= \phi_0(1 + \epsilon_1), \\ \phi^*(\delta) &= \phi(\delta)(1 + \epsilon_2), \end{aligned} \qquad |\epsilon_1|, |\epsilon_2| \leq \eta.$$

If $\phi_0 \simeq \phi(\delta)$, then

$$\Delta\phi^* = (\phi^*(\delta) - \phi_0^*) \simeq \Delta\phi + \phi_0\epsilon_3, \qquad |\epsilon_3| \leq 2\eta.$$

Then the relative error due to cancellation is

$$\left|\frac{\Delta\phi^* - \Delta\phi}{\Delta\phi}\right| \simeq \left|\frac{\phi_0\epsilon_3}{\Delta\phi}\right| \leq 2\left|\frac{\phi_0}{\Delta\phi}\right|\eta. \tag{9}$$

Comparing (8) and (9), we see that if $\Delta\phi$ increases with δ the effect of cancellation error is dominant for small δ and decreases as δ increases until the effect of truncation error, which is negligible for small δ, becomes dominant. Thus a good choice for δ should be that value which causes the bound (9) to be equal to (8). If there is more than one such δ, (8) indicates that the smallest one should be chosen. Thus δ is required to be the smallest number satisfying

$$|\alpha||\delta||\Delta\phi| - 4|\phi_0||\gamma|\eta = 0.$$

The case where $\alpha, \gamma > 0$ is typical (indeed in the Davidon method α, which is estimated by $\alpha_{jj}^{(i)}$, is always positive). If $\delta_p = |\delta|$, then the above equation may be broken up into

$$\tfrac{1}{2}\alpha^2\delta_p^3 + |\alpha||\gamma|\delta_p^2 - 4|\phi_0||\gamma|\eta = 0, \qquad \delta > 0, \tag{10a}$$

$$-\tfrac{1}{2}\alpha^2\delta_p^3 + |\alpha||\gamma|\delta_p^2 - 4|\phi_0||\gamma|\eta = 0, \qquad -2|\gamma|/|\alpha| \leq \delta \leq 0, \tag{10b}$$

$$\tfrac{1}{2}\alpha^2\delta_p^3 - |\alpha||\gamma|\delta_p^2 - 4|\phi_0||\gamma|\eta = 0, \qquad \delta \leq -2|\gamma|/|\alpha|. \tag{10c}$$

Then δ_p should be taken as the smallest positive root of (10a), (10b) or (10c). However, the single positive root of (10a) is smaller than any positive root of the other two equations, so this is the required δ_p. More generally for all signs of α and γ, δ_p should be calculated from (10a), and δ given the same sign as γ if α is positive and the opposite sign if α is negative. This is equivalent to saying that in approximating the derivative of a parabola by simple differences the direction in which the difference is taken should point away from the stationary point of the parabola.

In practice the following procedure has been found to give an adequate approximation to the positive root of (10a). Depending on which will give the smallest value, drop the cubic or quadratic term from (10a) and solve the resulting polynomial. That is,

$$\delta_p' = 2\left\{\frac{|\phi_0|}{|\alpha|}\right\}^{\frac{1}{2}}, \qquad \gamma^2 \geq |\alpha||\phi_0|\eta, \tag{11a}$$

or

$$\delta_p' = 2\left\{\frac{|\phi_0||\gamma|}{\alpha^2}\right\}^{\frac{1}{3}}, \qquad \gamma^2 \leq |\alpha||\phi_0|\eta. \tag{11b}$$

Refine this approximation by one application of Newton's method:

$$\delta_p \simeq \delta_p'\left[1 - \frac{|\alpha|\delta_p'}{3|\alpha|\delta_p' + 4|\gamma|}\right], \tag{12a}$$

or

$$\delta_p \simeq \delta_p'\left[1 - \frac{2|\gamma|}{3|\alpha|\delta_p' + 4|\gamma|}\right]. \tag{12b}$$

The above equations require that some estimate η_ϕ of the error bound on the calculation of ϕ be supplied. However a lower error bound may be gotten from the fact that the function evaluations take place not around zero, but around some point ξ which may be represented only approximately in the machine by ξ^*. If $\xi^* = (1 + \epsilon)\xi$, where $|\epsilon| < \eta_m$, then

$$\phi(\xi^*) \simeq \phi(\xi) + \gamma\epsilon\xi,$$

where γ now equals $\phi'(\xi)$. Thus

$$\phi(\xi^*) \simeq \phi(\xi)(1 + \epsilon'), \qquad |\epsilon'| \leq \left|\frac{\gamma\xi}{\phi(\xi)}\right| \eta_m.$$

If $|\gamma\xi/\phi(\xi)|\,\eta_m > \eta_\phi$, this larger number should be taken as η in the above.

Finally if, after δ has been computed, the value in eq. (8) is greater than some prescribed upper bound, say 10^{-m}, then the derivatives may not be sufficiently accurate and it is necessary to use some higher order scheme. For the modified Davidon method central differences are used:

$$\gamma \simeq \frac{(\phi(\xi + \delta) - \phi(\xi - \delta))}{2\delta}. \tag{13}$$

The truncation error in (13) depends on the nonquadratic properties of ϕ, which are unknown. The bound on the relative cancellation error in computing (13) is still given by (9) where $\Delta\phi$ now is equal to $\phi(\xi + \delta) - \phi(\xi - \delta)$, and it is possible to choose δ so that this error is less than 10^{-m}. The results of such an analysis are that δ should be equal to $10^{-m}\eta\,|\phi_0|/|\gamma|$; however, numerical experiments with noisy functions indicate that if η is relatively large and γ small, the resulting δ may be larger than the range in which the quadratic approximation may be reasonably expected to hold. An alternative which seems to work satisfactorily is to choose δ so that the cancellation error in $\phi(\xi + \delta) - \phi_0$ is less than 2×10^{-m} when $\text{sign}(\delta) = \text{sign}(\gamma)$. Thus δ is chosen as the positive root of

$$\tfrac{1}{2}|\alpha|\delta^2 + |\gamma|\delta - 10^m|\phi_0|\eta = 0. \tag{14}$$

The results of this section may be summarized in the following procedure, in which it is assumed that estimates are available for α, γ and $\phi_0 = \phi(\xi)$.

$$\left.\begin{array}{l}
\text{1. Set } \eta = \max\left\{\eta_\phi,\ \dfrac{|\gamma|}{|\phi_0|}\,|\xi|\,\eta_m\right\}. \\
\text{2. Calculate } \delta_p \text{ from (11) and (12) and set } \delta = \text{sign}(\alpha)\,\text{sign}(\gamma)\delta_p. \\
\text{3. If } .5\,|\alpha\delta/\gamma| < 10^{-m}\text{, estimate } \gamma \text{ from (7). Otherwise recompute } \delta \\
\quad\ \text{from (14) and use (13).}
\end{array}\right\} \tag{15}$$

The analysis in this section is only approximate and ignores some important cases. First, it assumes that ϕ may be represented by a parabola over the range within which differences are to be taken. Second, the justification for determining δ by requiring the equality of (8) and (9) is on somewhat shaky ground since the two sources of error may interact; however, this interaction can only occur when the two sources of error are approximately equal, so that this probably yields a fair approximation of the δ which actually minimizes the error. Third, the relative error formula (8) breaks down when $\gamma = 0$. Fourth, the analysis assumes that a constant error bound is known for ϕ which holds at all points at which it might be evaluated. Finally, no attempt has been made to justify the use of estimates for α, γ and ϕ_0 obtained from Davidon's method on other than heuristic grounds. On the other hand the numerical results to be quoted later indicate that the procedure (15) is effective for a wide variety of functions.

4. *Programming Details*

The modified Davidon method as outlined above has been coded in FORTRAN IV for the IBM 7090. It consists of using the procedure given above for computing derivatives at step 3 to compute the gradient vector g_{i+1}. As suggested $\Phi(x_{i+1})$, $\gamma_j^{(i)}$ and $\alpha_{jj}^{(i)}$ replace ϕ_0, γ and α in (15) in the computation of the jth component of g_{i+1}. The $\alpha_{jj}^{(i)}$ are carried along in the iteration by means of the modification rule (4). If the relative truncation error (8) is greater than 10^{-2}, then central differences are used to calculate the derivatives. Since the first iteration requires an initial gradient calculation and a second gradient calculation for which there are no estimates of the second derivatives, the user must supply a vector of appropriate step sizes for the calculation of the first two gradients by simple differences. The input to the subroutine is as follows:

An initial point x_0.
The number of variables n.
A lower bound on Φ, used in the linear minimizations.
A vector e specifying the final accuracy required of the components of x.
A vector of step sizes used in the initial approximation of the gradient.
The name of a user coded subroutine to evaluate the function.
An estimate of the relative error in the function evaluations, η_ϕ.
An upper bound on the number of linear minimizations to be performed by the program.
A parameter controlling the amount of information written out in the course of the iteration.

As suggested by Fletcher and Powell, the iteration is terminated when each of the components of d_i and $\sigma_i d_i$ is less than the required accuracy specified by the components of e. The program also returns if the number of linear minimizations has exceeded the specified upper bound, if a linear minimization has failed to change the position or the function value, or if the square of the magnitude of the gradient underflows the machine. The usual cause of a return through one of these exists is overly stringent convergence criteria.

The matrices H_i must be positive definite. In particular this means that the diagonal elements of H_i^{-1}, the $\alpha_{jj}^{(i)}$, must be positive and that the derivative of the function along the predicted direction, $g_i^T d_i$, must be negative. If either of these conditions is violated, the iteration is restarted by setting H_i equal to the identity matrix. If the step size σ_i along d_i is negative, the H_{i+1} generated by (7) may not be positive definite. In this case the iteration is also restarted with the identity matrix.

The procedure for minimizing along a line is a variation of the method outlined in Powell [6] with the following differences:

1. The initial step size along d_i is chosen as

$$\min\left\{1, \frac{-2(\Phi(x_i) - \Phi_{\text{lower}})}{g_i^T d_i}\right\},$$

TABLE 1.
A PARABOLIC VALLEY

Iteration No.	*Number of function evaluations*	Φ
0	1	2.4 10^{1}
2	13	3.8 10^{0}
4	26	2.9 10^{0}
6	39	1.9 10^{0}
8	56	7.1 10^{-1}
10	70	2.9 10^{-1}
12	83	1.4 10^{-1}
14	97	5.4 10^{-2}
16	111	1.8 10^{-2}
18	124	1.3 10^{-3}
20	132	1.7 10^{-6}
22	145	2.8 10^{-10}
23	152	1.0 10^{-11}
24	163	9.0 10^{-12}
25	169	3.3 10^{-12}
26	174	3.3 10^{-12}

TABLE 2.
A HELICAL VALLEY

Iteration No.	*Number of function evaluations*	Φ
0	1	2.5 10^{3}
2	9	5.2 10^{2}
4	28	2.5 10^{1}
6	39	1.1 10^{1}
8	54	7.2 10^{0}
10	72	1.7 10^{0}
12	85	5.8 10^{-1}
14	99	1.1 10^{-2}
16	109	8.8 10^{-4}
18	119	1.7 10^{-8}
20	130	1.1 10^{-12}
21	136	6.7 10^{-16}

TABLE 3.
A FUNCTION OF FOUR VARIABLES

Iteration No.	*Number of function evaluations*	Φ
0	1	2.2 10^{2}
2	21	1.9 10^{1}
4	37	5.5 10^{-2}
6	54	1.1 10^{-2}
8	69	3.4 10^{-3}
10	83	2.7 10^{-3}
12	104	2.8 10^{-5}
14	122	1.8 10^{-7}
16	139	1.1 10^{-8}
18	158	8.8 10^{-9}
19	168	8.8 10^{-9}
⋮	⋮	⋮
41	407	1.1 10^{-10}

where Φ_{lower} is the lower bound on Φ provided as input to the program. This choice is recommended in Fletcher and Powell.

2. A second estimate of the minimum is chosen by parabolic interpolation on the function value at x_i, the function value at the first step and the derivative of the function along d_i at x_i, $g_i^T d_i$.

Once these three function values have been determined the process can proceed by parabolic interpolation or extrapolation to the minimum. If an extrapolation is performed, its distance from the nearest endpoint is required to be less than twice the distance of that endpoint from the central point.

5. *Numerical Results and Discussion*

The program outlined in Section 4 has been used to minimize a variety of test functions which appear frequently in the literature. With one exception (Chebyquad), these functions are designed to be difficult to minimize, and with the exception of functions occurring in least squares problems they are not representative of the type of functions which commonly appear in applications. However, they serve well to point out specific shortcomings in a particular method and, as Fletcher has pointed out in his review [2], they also serve as a standard of comparison among methods. The results quoted about Powell's method are from Fletcher's review.

All the functions presented were computed in single precision on an IBM 7090 computer. In the description of the functions that follows, an attempt is made to indicate the manner in which they were computed so that the reader may attempt to determine the final accuracy of the function values at various points. The input convergence criterion was 10^{-6} for all components and the estimated relative accuracy of the function evaluations was 10^{-8}. The tables record the progress of the method for each test function, giving the iteration number, the number of function evaluations and the function value at the end of each iteration.

TABLE 4. CHEBYQUAD, $n = 2$

Iteration No.	Number of function evaluations	Φ	Number of function evaluations	$\Phi - 1$	Number of function evaluations	$\Phi - 1$
0	1	$1.2 \quad 10^{-1}$	1	$1.2 \quad 10^{-1}$	1	$1.2 \quad 10^{-1}$
1	7	$2.1 \quad 10^{-6}$	7	$2.1 \quad 10^{-6}$	10	$2.1 \quad 10^{-4}$
2	12	$8.2 \quad 10^{-7}$	13	$7.0 \quad 10^{-7}$	15	$8.5 \quad 10^{-6}$
3	16	$2.1 \quad 10^{-8}$	20	0.0	22	$3.3 \quad 10^{-6}$
4	20	$9.4 \quad 10^{-11}$	27	0.0	28	$-8.5 \quad 10^{-6}$
5	25	$2.3 \quad 10^{-12}$	49	0.0	39	$-8.5 \quad 10^{-6}$
6	31	$1.7 \quad 10^{-14}$				

(a) *A Parabolic Valley* (Rosenbrock, Table 1):

$$\Phi(x) = 100(\xi_2 - \xi_1^2)^2 - (1 - \xi_1)^2$$
$$x_0 = (1.2, 1.0) \qquad \text{(starting point)}$$
$$\Phi(1, 1) = 0 \qquad \text{(minimum).}$$

This function requires the method to follow a steep-sided parabolic valley to the minimum at (1, 1). After 163 function evaluations the method had produced an $x = (1.000002, 1.000003)$ with a function value of 9×10^{-12}. The method then broke down and stopped after two more iterations because it was unable to reduce the function value along the predicted direction. This breakdown was due to the failure of (15) to produce accurate derivatives around the minimum. Up to the point of failure, the method compared well with Powell's method.

(b) *A Helical Valley* (Fletcher and Powell, Table 2):

$$\Phi(x) = 100[(\xi_3 - 100)^2 + (\rho - 1)^2] + \xi_3^2$$
$$2\pi\theta = \tan^{-1}(\xi_2/\xi_1) \qquad (-.25 < \theta < .75)$$
$$\rho = (\xi_1^2 + \xi_2^2)^{\frac{1}{2}}$$
$$x_0 = (-1, 0, 0)$$
$$\Phi(1, 0, 0) = 0.$$

This function poses a valley following problem in three dimensions. The method converged after 136 function evaluations with a minimum value of 7×10^{-16}.

(c) *A Function of Four Variables* (Powell, Table 3):

$$\Phi(x) = (\xi_1 + 10\xi_2)^2 + 5(\xi_3 - \xi_4)^2 + (\xi_2 - 2\xi_3)^4 + 10(\xi_1 - \xi_4)^4$$
$$x_0 = (3, -1, 0, 1)$$
$$\Phi(0, 0, 0, 0) = 0.$$

The matrix of second derivatives of this function at its minimum is singular, thus making it a particularly difficult test for quadratically convergent methods. As with Rosenbrock's parabolic valley, the modified Davidon method broke down (after the 18th iteration); however, in this case the breakdown was caused by cancellation in the computation of $d_i = -H_i g_i$, for which the only cure would be to calculate the gradient to excessive accuracy. Throughout the iteration the scheme (25) pro-

TABLE 5. CHEBYQUAD, $n = 4$

Iteration No.	Number of function evaluations	Φ	Number of function evaluations	$\Phi - 1$	Number of function evaluations	$\Phi - 1$
1	1	7.1 10^{-2}	1	7.1 10^{-2}	1	7.1 10^{-2}
2	8	1.2 10^{-2}	8	1.2 10^{-2}	12	2.6 10^{-2}
3	16	8.0 10^{-3}	16	8.0 10^{-3}	23	1.0 10^{-2}
4	23	6.2 10^{-4}	23	5.5 10^{-4}	33	8.5 10^{-3}
5	29	2.9 10^{-4}	30	2.8 10^{-4}	45	1.8 10^{-3}
6	35	1.9 10^{-4}	39	2.3 10^{-4}	57	1.0 10^{-3}
7	41	1.6 10^{-7}	45	3.0 10^{-7}	68	2.3 10^{-5}
8	47	1.3 10^{-9}	54	0.0	82	2.3 10^{-6}
9	55	8.1 10^{-12}	71	0.0	94	-6.3 10^{-7}
10	65	2.6 10^{-14}			133	-6.3 10^{-7}

duced derivatives accurate to the required two significant figures. The method continued for 23 more iterations, stopping when the convergence criteria were satisfied. This in fact represents a failure of the method for determining convergence, since the components of the final point were only of order 10^{-3}. Before the breakdown the method stayed well ahead of Powell's method, which required 235 function evaluations to reach 5×10^{-9}, and the method of Davies, Swann and Campey, which required 180 evaluations to reach 10^{-8} (but note that both methods continued to reduce the function at a satisfactory rate).

(d) *Chebyquad* (Fletcher, Tables 4, 5 and 6). The computation was done by a straightforward FORTRAN implementation of the ALGOL procedure in Fletcher [2].

$$x_0 = \left(\frac{1}{n+1}, \frac{2}{n+1}, \cdots, \frac{n}{n+1}\right).$$

This function, devised by Fletcher, is more typical of the functions found in applications. It was run for $n = 2$, 4 and 6, for which the function has a minimum of zero. In all cases the method and the convergence criterion were entirely satisfactory. For $n = 6$, Powell's method required 288 function evaluations to reach 7×10^{-14}. Two additional tests were performed with these functions and the results are given in the additional columns in the tables. First the function values were increased by one to test the ability of the method to converge to a nonzero minimum. The results indicate that the course of the iteration was not much affected. For this test the convergence criteria could not be met and the method stopped when an iteration failed to decrease the function. In the second test the function values were increased by one and then given a relative perturbation of about 10^{-5} using a pseudo-random-number generator. The method required more function evaluations to each a minimum, but the convergence was quite satisfactory.

Additional tests were run on the trigonometric functions of Fletcher and Powell [3]. The parameters in these functions may be chosen randomly so that, for given n, sets of functions with known minima may be generated. The function values were computed in double precision and then rounded off to single precision. A final accuracy of 10^{-4} was required in all components of x and the relative error in the function evaluations was estimated at 10^{-3}. Cases were run for $n = 5$, 10 and 20 and the results are summarized in Table 7. In every case the method converged to the proper minimum. Powell [6] minimized these functions and a comparison of Table 7 with his results suggests that for $n = 20$ the modified Davidon method requires between one-half to one-third the number of function evaluations to attain the same accuracy as Powell's method.

The results of these tests indicate that the modified Davidon method is an efficient technique for minimizing most functions of several variables for which no derivatives are readily available. The breakdown of the method on Rosenbrock's parabolic valley and Powell's function of four variables raises the possibility that the method might also fail in other cases if it is pressed too far. However, it should be noted that with the parabolic valley the method had virtually attained the desired accuracy before stopping and that for both functions the initial progress of the iteration was very good. The only function for which the convergence criterion stopped the iteration before the desired accuracy was attained was Powell's function of four variables.

In terms of number of function evaluations, the modified Davidon method seems

TABLE 6. CHEBYQUAD, $n = 6$

Iteration No.	*Number of function evaluations*	Φ	*Number of function evaluations*	$\Phi - 1$	*Number of function evaluations*	$\Phi - 1$
0	1	4.6 10^{-2}	1	4.6 10^{-2}	1	4.6 10^{-2}
1	10	2.8 10^{-2}	10	2.8 10^{-2}	10	2.9 10^{-2}
2	20	1.8 10^{-2}	20	1.8 10^{-2}	22	2.1 10^{-2}
3	31	1.1 10^{-2}	31	1.1 10^{-2}	39	1.4 10^{-2}
4	40	9.2 10^{-3}	41	8.6 10^{-3}	56	1.2 10^{-2}
5	50	5.2 10^{-3}	50	5.2 10^{-3}	71	1.1 10^{-2}
6	58	1.6 10^{-3}	58	1.8 10^{-3}	86	4.1 10^{-3}
7	66	5.5 10^{-5}	66	6.1 10^{-5}	101	1.2 10^{-3}
8	74	2.0 10^{-5}	75	1.8 10^{-5}	115	5.6 10^{-5}
9	82	1.6 10^{-5}	89	1.6 10^{-5}	131	3.0 10^{-6}
10	93	4.8 10^{-6}	105	5.0 10^{-7}	149	-5.7 10^{-6}
11	103	9.6 10^{-9}	121	0.0	170	-5.7 10^{-6}
12	112	2.0 10^{-10}	143	0.0		
13	126	2.2 10^{-12}				
14	140	2.6 10^{-15}				

TABLE 7. TRIGONOMETRIC FUNCTIONS OF SEVERAL VARIABLES

n	*Number of function evaluations*	Φ
5	113	3.4 10^{-10}
5	63	2.7 10^{-7}
10	221	8.2 10^{-8}
10	291	9.3 10^{-9}
20	836	5.8 10^{-9}
20	689	6.8 10^{-7}
20	745	1.4 10^{-8}
20	649	6.0 10^{-8}

to be more efficient than Powell's, with the disparity becoming more marked for large n. However, in order to avoid linear dependence among the directions along which the function must be minimized, Powell has incorporated into his method a rather complicated criterion to determine whether a new direction should be accepted or the old directions used in the next iteration. The result of this is that as n increases new directions are chosen less often. If, as Fletcher suggests, this criterion is overly stringent and a better one could be found, then Powell's method might well be superior to the one proposed here. Nonetheless, the modified Davidon method is significantly more efficient than Powell's method as it now stands, and in addition it provides a natural and effective means for determining convergence.

Since the mechanics of the proposed method are somewhat time-consuming, Powell's method is probably more efficient in cases where function evaluations consume only a small amount of time. If the function to be minimized can be written as a sum of squares, then either the generalized least squares method, which requires derivatives, or a new method by Powell [7], which does not, is more efficient. If the gradient can be easily calculated, then Davidon's original method should be used. Otherwise, it is hoped that the method proposed here will provide an efficient technique for minimizing a differentiable function of several variables.

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